

Hannah Brake

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 portfolio |  linkedin |  github

EDUCATION

B. Eng. Computer Engineering

Memorial University of Newfoundland and Labrador

Sept. 2017 - Apr. 2022

GPA: 3.77/4.0 (83% Avg)

EXPERIENCE

Software Test Engineer

Jun. 2022 - Present

Nokia

Ottawa, Ontario

- Develop **Python/Pytest** automation and hardware test benches in a **Linux** regression environment, executing **system-level** tests on SROS and SRLinux-based Nokia 7750, 7250, and 7730 routers.
- Own **end-to-end test cycles** for Precision Time Protocol (PTP) hardware and software features, including reviewing **requirements** and developing **comprehensive test plans**.
- Verify synchronization timing distribution across multiple system **FPGAs** and **ASICs**, and validate C++ application software features.
- **Triage** regression failures, **debug** root causes with software and hardware teams, file defects, and verify fixes.
- Screen, **interview**, and help select software test co-ops and full-time hires; **train and mentor** new team members on tools, environment, and process.
- Represent Nokia at the **University of Ottawa** and **Kanata North career fairs** (2023), speaking with 200+ students about 5G and software testing.

Mission Control System Developer

Jun. 2021 - Apr. 2022

C-CORE, Memorial University of Newfoundland

St. John's, Newfoundland

- Worked on a team of five for a Canadian Space Agency initiative to develop **I2C-based** mission control software for the Memorial University **Killick-1 CubeSat**.
- Wrote an API in **C** to manage and monitor voltage rails and power distribution to subsystem hardware modules. Developed **platform test** scripts and ran **hardware-in-the-loop testing** on a custom flat-sat board.
- Developed with **Code Composer Studio**, the ABACUS Satellite On-Board Computer, and the ISISPACE Compact Electrical Power System.
- Wrote **technical documentation** of hardware/software testing, C development, tools, and future CubeSat plans. Presented to Canadian Space Agency representatives.

Embedded Systems Developer Co-op

Sept. 2019 - Dec. 2019

Makerspace, Memorial University of Newfoundland

St. John's, Newfoundland

- Built a **Raspberry Pi 3** security-checkpoint prototype that scanned university ID cards, logged access, and gated entry, using a 16x2 **GPIO LCD**, **multi-threaded Python**, and **MySQL**.

PROJECTS

macOS 27 Personalization (Notion | Xcode | Swift)

Built native-feeling **macOS 27** apps for desktop features that felt missing, while keeping the system UI intact. Planned in **Notion** with a database of features (documented as completed, needing maintenance, and future).

- **Welcome Island**  — On laptop unlock, an island expands around the camera notch, cycling through a personal greeting, Calendar events, and suggested app activities.
- **Window Navigation Panel**  — Glassy left-edge panel of open-app icons; nudge the cursor to the leftmost part of the screen to switch instantly between open app windows without the noise of using the built-in Dock.
- **Menu Bar Resource Monitor**  — View live CPU, memory, and network utilization metrics from the menu bar, plus a floating desktop panel widget.

Penga (Xcode | Swift)

- A friendly penguin virtual pet that lives in the iPhone **Dynamic Island**. Dynamic Island and **Live Activities** show her state, including happiness metrics and pixel art. She can automatically listen to music with you!

Dottie (Raspberry Pi 5 | LineageOS)

- Custom **Raspberry Pi 5**-based device with a circular Waveshare capacitive touch display, designed to sit with a home vinyl, turntable, and speaker setup. The Pi runs Android-based **LineageOS**. Connects to Bluetooth speakers and shows album art and track info while listening.

SKILLS

Programming Languages Python | TCL | Bash | MATLAB | C++/C | C# | VHDL | Assembly | Java | Swift

Tools Oscilloscope | Raspberry Pi | Arduino IDE | VS Code | Xcode | Cursor | Notion | MobaXterm | Confluence | Docker

Technologies Linux | Git | GitLab | CI/CD pipelines | FreeRTOS | Altera FPGA | ModelSim | Pytest | MySQL | Wireshark